3a. Design and implement C/C++ Program to solve All-Pairs Shortest Paths problem using Floyd's algorithm.

```
#include<stdio.h>
int a[10][10],i,j,k,n;
int min(int,int);
void floyd(int a[10][10],int n)
 {
for (int k=1; k<=n; k++)
{
   for (int i=1; i<=n; i++)
   for (int j=1; j<=n; j++)
   a[i][j] = min(a[i][j], a[i][k] + a[k][j]);
 }
printf("All Pairs Shortest Path is :\n");
for(int i=1;i<=n;i++)
{
    for(int j=1;j<=n;j++)
       {
          printf("%d ",a[i][j]);
       }
        printf("\n");
 }

}
        int min(int a,int b)
         {
          if(a>b)
         return b;
        else
          return a;
          }

        int main()
        {
          printf("\nEnter the no. of vertices:");
          scanf("%d", &n);
          printf("\nEnter the cost adjacency matrix:\n");
         for (i = 1; i<= n; i++)
        {
         for (j = 1; j<= n; j++)
```

```
 {
          scanf("%d", &a[i][j]);
         }
 }
floyd(a,n);
}
```

OUTPUT

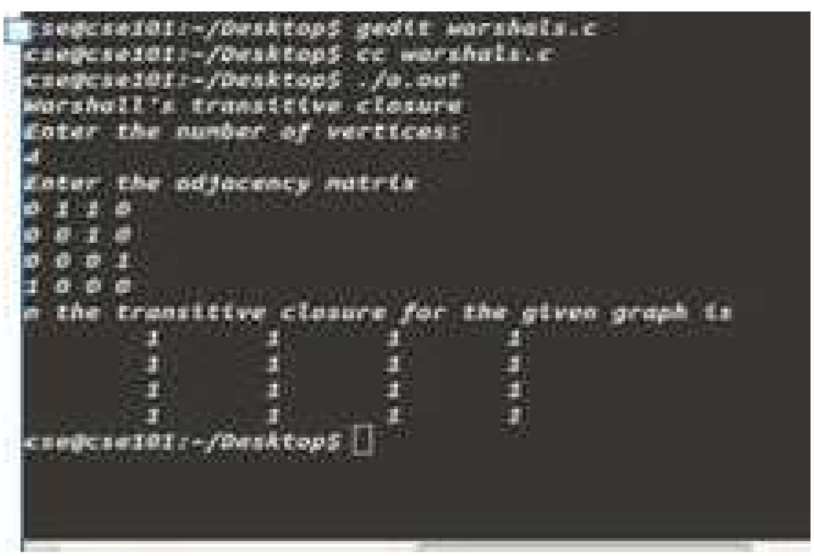
3b. Design and implement C/C++ Program to find the transitive closure using Warshal's algorithm.

```
#include<stdio.h>
int a[10][10],n;
void warshall();
void main()
{
      int i, j;
      printf("Warshall's Transitive Closure\n");
      printf("Enter the number of vertices :\n");
      scanf("%d",&n);
      printf("Enter the adjacency matrix\n");
      for (i=1; i<=n; i++)
            for (j=1; j<=n; j++)
                  scanf("%d",&a[i][j]);
      warshall();
      printf("nThe transitive closure for the given graph is\n");
      for (i=1; i<=n; i++)
      {
```

```
        for (j=1; j<=n; j++)
        {
            printf("%d",a[i][j]);
        }
        printf("\n");
    }

    }
    void warshall()
    {
    int i,j,k;
    for (k=1; k<=n; k++)
    {
        for (i=1; i<=n; i++)
        {
            for (j=1; j<=n; j++)
            {
                if (a[i][j] || (a[i][k] && a[k][j]))
                    a[i][j] = 1;
            }
        }
    }
}
```

OUTPUT



```
cse@cse101:~/Desktop$ gedit warshals.c
cse@cse101:~/Desktop$ cc warshals.c
cse@cse101:~/Desktop$ ./a.out
Warshall's transitive closure
Enter the number of vertices:
4
Enter the adjacency matrix
0 1 1 0
0 0 1 0
0 0 0 1
1 0 0 0
n the transitive closure for the given graph is
        1       1       1       1
        1       1       1       1
        1       1       1       1
        1       1       1       1
cse@cse101:~/Desktop$
```